



Armada Kalamunda Group

Anaemia in Pregnancy Guideline

1. Purpose

To Identify and treat women with iron deficiency and iron deficiency anaemia (IDA), to improve maternal, fetal and newborn outcomes.

Women with anaemia and/or iron deficiency may experience fatigue, reduced energy levels and reduced mental performance. Severe anaemia is associated with preterm birth, low birth weight, and a small for gestational age fetus. In the postpartum period both anaemia and iron deficiency have been found to be linked to depression, emotional instability, stress, lower cognitive performance tests, poor tissue healing and increased risk of infections.

2. Applicability

This guideline applies to the midwifery, nursing and medical disciplines within the Obstetrics and Gynaecology departments of Armada Health Service.

3. Guidelines

Anaemia in pregnancy is a global health problem. While some degree of dilutional anaemia is part of normal pregnancy physiology, iron deficiency anaemia can have serious adverse health consequences for the mother and child. Thus, it is critical to distinguish iron deficiency anaemia from physiologic anaemia, as well as to identify other less common causes of anaemia that may require treatment. The overwhelming majority of anaemia in reproductive-age women is due to low or absent iron stores, making iron deficiency anaemia the world's most common anaemia due to a micronutrient deficiency. The prevalence of iron deficiency increased from 7 percent in the first trimester to 24 and 39 percent in the second and third trimesters, respectively. Women with iron deficiency may be at risk for developing iron deficiency anaemia during pregnancy, when the demand for iron increases.

3.1 Causes of anaemia

- Physiological
- Iron deficiency is a common cause of anaemia in pregnancy, in both the developed and developing world.
- Other causes of anaemia include megaloblastic anaemias due to vitamin B12 and folic acid deficiency, Thalassaemia's, blood loss, haemolytic states (Sickle Cell Disease, malaria and pre-eclampsia), helminthic infection and underlying malignancy and chronic disease.

The treatment of anaemia requires an accurate assessment of its underlying cause, but this can be difficult in pregnancy, where multiple factors may be responsible.

The preferred test of maternal iron status is the serum ferritin level, although, because ferritin is an acute phase reactant, levels can be elevated in inflammatory states

Other measures of iron status (e.g. serum iron, transferrin, transferrin receptors and erythrocyte protoporphyrin) have a limited role in pregnancy, due to restricted availability of tests, cost and interpretive difficulties arising from non-standardised reference ranges and diurnal variation.

3.2 Key points

- All women should be offered screening for anaemia:
 - in first trimester (or at booking) (Full Blood Picture (FBP) and iron studies) irrespective of the FBP and iron levels at the time of referral
 - with the next screening bloods (usually performed between 24-28 weeks) see Treatment algorithm for tests required
 - and at 36 weeks gestation
- A FBP should be ordered within 2 - 8 weeks following initiation of treatment (dependent upon gestation) to assess response and compliance with oral iron treatments.
- Iron deficiency in most circumstances is diagnosed by a full blood examination and serum ferritin levels. Do not use serum iron, or serum ferritin alone to diagnose iron deficiency.

NB: Ferritin levels are elevated in active infection or inflammation and in these cases measurement of C-reactive protein (CRP) will support interpretation of ferritin levels.

- Consider consulting a Haematologist in diagnosing and treating Iron Deficiency Anaemia (IDA) in women with known haemoglobinopathies. Serum ferritin should be checked prior to starting iron with known haemoglobinopathy.
- Oral iron if taken at the appropriate dose, and for enough time, is an effective first-line treatment for most women in pregnancy.
- If a woman fails to respond to iron therapy, investigate further to assess for malabsorption problems. Consider non-compliance with medications or co-existing disease.
- Intravenous iron therapy is an effective alternative to oral treatment during the second or third trimester only for treatment of IDA. Intravenous iron is restricted to women failing to respond to oral iron treatment with known IDA or in those in whom a rapid repletion of ferritin is required.
- The type, dosage, and frequency of iron supplements for treatment of IDA should be documented 'Pregnancy Health Record'.
- At each antenatal visit all women taking iron supplements should be monitored for medication compliance and side-effects.

3.3 Prescribing iron supplements and follow-up:

- All women with known haematinic deficiencies (includes B12 or iron deficiency) should be advised:
 - the type, frequency, and duration of the treatment or medication
 - side-effects of the medication which can exacerbate the symptoms of pregnancy including heartburn, nausea, vomiting and constipation.
 - management of side-effects
 - how and when to take the medications
 - of medications or food that may inhibit iron absorption
 - dietary information to increase oral iron intake.
- Provide written instructions to the woman about iron supplementation.
- At each antenatal visit:
 - assess and document the women for compliance with taking the medication.

- assess and document side-effects from the medication. Provide advice for management of any side-effects.
- assess compliance to dietary recommendations.

3.4 Recommendations for prescribing oral and/or parenteral iron: (Appendix 1)

- The routine administration of iron supplementation to all pregnant women is not recommended.
- In maternity patients who require iron therapy for the treatment of anaemia, the routine addition of folic acid is not recommended.
- In maternity patients with iron deficiency anaemia, a therapeutic dose of elemental iron (100–200 mg daily) should be prescribed, and the response to therapy monitored. If the response to oral iron is inadequate, IV iron should be used.
- In maternity patients with iron deficiency without anaemia, a low dose of elemental iron (e.g. 60–80 mg daily) may be considered, and may be better tolerated than higher doses.
- In maternity patients requiring iron, IV iron is preferred when oral iron is poorly tolerated (affecting compliance), or absorption is likely to be impaired.
- Iron infusion process for maternity patients:
 - In non-urgent situations, GP Obstetrician or midwife creates a phone referral to Haematology on 93912673, Tuesday and Friday. Relevant documentation must be given to Haematology staff or faxed prior to the day of appointment on 93912707. Refer to the [Iron infusion-Outpatient Process Procedure](#)
 - In urgent situations generally in late third trimester where Haematology cannot accommodate the patient, an appointment is scheduled in Antenatal Assessment Unit (AAU) for iron infusion. The GP Obstetrician gives a script to the woman and all other documents are faxed or send in person to AAU for preparation of iron infusion. Generally, Ferrinject (Ferric Carboxymaltose) short infusion is preferred choice. However, in some situations Ferrosig (Iron Polymaltose) can be administered. The hospital pharmacy's preferred iron infusion is Ferrosig (Iron Polymaltose) long infusion.
- When IV iron is prescribed, calculation of the dose should take into consideration the iron deficit. Refer to the [Intravenous Iron Therapy Adult Guideline](#).
- The routine use of IM iron is not advised where alternatives are available.

3.4.1 Side-effects of oral medications and management:

When oral liquid iron is used it should be diluted with water and a straw used to prevent discolouration of the teeth. However, liquid iron supplements should be checked for the content of elemental iron.

Side-effects of oral iron supplements include nausea, epigastric pain, constipation and black discolouration of the faeces.

Management for side effects include:

- nausea and epigastric discomfort – take iron tablets on an empty stomach 1 hour prior to or 2 hours after a meal, commence tablets on a low dosage and then gradually increase the amount of iron, or take small doses more frequently.
- constipation

3.5 Dietary information

Sources of dietary iron include meat, poultry and fish which are two to three times more absorbable than plant-based iron foods. Iron-fortified foods and ascorbic acid enhances absorption. Vegetarians should be encouraged to eat foods high in iron, such as, tofu, beans, lentils, spinach, whole wheat breads, peas, dried apricots, prunes and raisins.

Medications inhibiting absorption or contraindicated include:

- anticonvulsants
- sulphonamides
- medications that raise gastric pH e.g. antacids (avoid where possible)

Foods that may interact or inhibit absorption:

- calcium in dairy products e.g.
- tea and coffee
- chocolate
- spinach and beetroot
- soy products
- phytates (salts found in plants capable of forming insoluble complexes with iron) e.g. bran, cereal.

Non-haem iron requires an acidic pH to be reduced to ferrous for gut absorption. A gap of two hours from dietary or medication inhibitors of iron absorption appears to be enough to avoid the problem.

3.6 Anaemia and haemoglobinopathies:

Haemoglobinopathies are autosomal recessive disorders which imply that they must be inherited through both parents who may be carriers and have the disorder themselves.

3.6.1 Populations at risk of Haemoglobin Disorders

Populations at risk for haemoglobin disorders include:

Thalassemia	Sickle cell disease
• African • American/British/Caribbean African • South East Asian and Chinese • Middle Eastern • Pacific Islanders • New Zealand Maori • Southern Europe/Mediterranean • Indian subcontinent • Some northern Western Australian and Northern Territory Australian indigenous communities.	• African • American/British/Caribbean African • Middle Eastern • Southern Europe/Mediterranean • Indian subcontinent • South American

3.6.2 Effect of Thalassaemia Types in Pregnancy:

Type of Thalassaemia	Genotype	Effect
α thalassaemia trait α^+ (α thalassaemia minor)	One deleted α gene in one chromosome only e.g. $-\alpha/\alpha\alpha$	Asymptomatic normally. Slight decrease in MCV/MCH
α thalassaemia trait α^+ (α thalassaemia minor)	Two deleted α genes, one from each chromosome e.g. α^-/α^-	Mild anaemia, slight decrease in MCV/MCH
α thalassaemia trait α^0 (α thalassaemia minor)	Deletion of two α genes in one chromosome only e.g. $--/\alpha\alpha$	Mild anaemia, decrease in MCV/MCH
Haemoglobin H disease	Total of three deleted α genes, in both chromosomes $\alpha^-/--$	Moderate anaemia, microcytic anaemia, hepatosplenomegaly, may require transfusion
Haemoglobin Barts hydrops	Complete absence of all α genes in both chromosomes $--/--$	Usually death in utero, hydrops fetalis syndrome
β thalassaemia trait (β thalassaemia minor)	One absent (0) or defective (+) β gene in one chromosome only e.g. β/β^0 or β/β^+	Asymptomatic normally. May have mild anaemia, normal/reduced MCV
beta thalassaemia intermedia	Two defective β genes or one absent and one defective in each chromosome e.g. β^+/β^+ or β^0/β^+	Variability in symptoms. Moderate anaemia, hepatosplenomegaly
β thalassaemia major	Two absent beta genes in both chromosomes β^0/β^0	Severe anaemia. Require frequent blood transfusions
Haemoglobin E trait (HbE)	One normal β gene in one chromosome and one abnormal E (β gene mutation) in the other chromosome e.g. β/β^E	Asymptomatic, possible microcytosis and decreased MCV/MCH
Haemoglobin E disease	Two abnormal E (β gene mutations), one in each chromosome β^E/β^E	Mild anaemia and microcytosis, decrease in MCV/MCH
Haemoglobin E/ β thalassaemia	Total one abnormal E (β gene mutation) and one absent (0) or defective (+) β gene in both chromosomes e.g. β^E/β^0 or β^E/β^+	Variable symptoms, moderate to severe anaemia, greater decrease in MCV/MCH. Thalassaemia intermedia or major

3.6.3 Effect of Sickle Cell in Pregnancy:

Sickle Status	Genotype	Effect
Sickle cell trait	One normal β gene in one chromosome and one abnormal S (β gene mutation) in the other chromosome e.g. β/β^S	Asymptomatic normally, normal RBC indices. Sickle cell formation can occur in during high fever and significant hypoxia
Sickle cell disease/anaemia	Two abnormal S (β gene mutations), one in each chromosome β^S/β^S	Mild to moderate chronic haemolytic anaemia, vaso-occlusion - brain, chest, bones, kidneys, spleen and placenta. Increased maternal and perinatal mortality

3.7 Screening Women for Haemoglobin Disorders: (Appendix2)

3.7.1 B12 deficiency and anaemia:

B12 deficiency is rare; particularly in pregnancy however B12 deficiency should be excluded in women with unexplained anaemia, or in women who fail to respond to treatment for iron deficiency anaemia. In normal pregnancy, B12 levels fall by 30% by the third trimester of pregnancy.

3.7.2 Women at risk of developing B12 deficiency:

- Elderly women (due to prevalence of gastric atrophy)
- Vegetarian diet, particularly vegan diets
- Previous gastric/ileac resection, or history of coeliac disease, inflammatory bowel disease
- Prolonged use of proton pump inhibitors, H2 receptor antagonists and
- Biguanides (may interfere with absorption of B12 over time)

3.7.3 Preventing and treating B12 deficiency

Dietary requirements for B12 are 2.4mcg/day for non-pregnant women. As B12 is animal sourced the recommendations for B12 supplements are for vegetarians and vegan women in pregnancy and lactation, with a recommended daily intake (RDI) 6 mcg/ day).

As the aetiology of B12 deficiency is generally absorptive, the recommended form of treatment is parenteral B12. If there is a strong suspicion of B12 deficiency, a short course of oral B12 should be given, with further investigation post-delivery by the patient's General Practitioner (GP).

Hydroxycobalamin or cyanocobalamin 1000mcg/1mL given by intramuscular injection, once weekly for 3 weeks. However, in severe cases of B12 deficiency, or if the patient is suffering from significant neurological symptoms, it can be administered more frequently. Seek Haematology advice if this is required.

It is important to correct any underlying iron deficiency, as treatment with B12 can cause rapid red cell production and associated iron depletion. Thus, it is important to assess ferritin levels and treat iron deficiency accordingly.

3.7.4 Notes on treatment of outpatients

The patient can arrange for treatment for B12 deficiency directly with her GP, or be provided with a prescription for 3 doses, to be administered at GP practice or on return visits to AKH.

3.7.5 Anaemia and folate deficiency

Folate is required for DNA synthesis; demand increases in pregnancy. Deficiency can develop rapidly as stores are minimal. The recommended dietary intake in pregnancy is 600µg/day, and most prescribed prenatal vitamins contain 800µg which is more than the recommended dose. Meat is not a good source for folate; however, folate can be found in green leafy vegetables, legumes and orange juice.

3.7.6 Management

Women at risk of folate deficiency (e.g. multiple pregnancy, haemolytic anaemia) should take 5 mg of folic acid throughout the pregnancy.

4. Legislative context

The following documents are required to give effect to this policy [i.e. the documents included are mandatory]:

- Health Department of Western Australia (HDWA) [Clinical Handover Policy MP 0095](#)
- HDWA [Consent to Treatment Policy OD 0657/16](#)
- [Medication Management Procedure](#)

5. Supporting information

The following documents support the implementation of this policy:

- [Recognition and Response to Acute Deterioration](#)
- [Iron Infusion – Outpatient Process Procedure](#)
- [Administration of Intravenous Iron Polymaltose for Haemodialysis Patients Procedure](#)

6. Compliance, monitoring and evaluation

Compliance to this guideline will be monitored through:

- Audit of women with Iron deficiency or Iron deficiency anaemia with regards to management of anaemia.
- Audit of IDA women requiring iron infusion and review of correct use of hospital Intravenous Iron therapy Adult guidelines

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 5 – Action 5.3: Clinicians use organisational processes from the Partnering with Consumers Standard when providing comprehensive care to:
- Standard 5 – Action 5.11: Clinicians comprehensively assess the conditions and risks identified through the screening process
- Standard 5 – Action 5.12: Clinicians document the findings of the screening and clinical assessment processes, including any relevant alerts, in the healthcare record
- Standard 5 – Action 5.13: Clinicians use processes for shared decision making to develop and document a comprehensive and individualised plan that:
- Standard 6 – Action 6.7: The health service organisation, in collaboration with clinicians, defines the:
- Standard 8 – Action 8.4: The health service organisation has processes for clinicians to detect acute physiological deterioration that require clinicians to:

8. References

1. World Health Organisation. Daily iron and folic acid supplementation during pregnancy. World Health Organisation 2017 http://www.who.int/elena/titles/daily_iron_pregnancy/en/
2. National Blood Authority (NBA). Patient blood management guidelines: Module 5 – Obstetrics and Maternity. NBA, Canberra, Australia. 2015.
3. Anaemia and iron deficiency: Management in pregnancy and postpartum, KEMH guideline
4. Patient Blood Management Guidelines: Module 5 Obstetrics and Gynaecology; <https://www.blood.gov.au/pbm-module-5>
5. Bain B J. Haemoglobinopathy diagnosis. 2nd ed. Carlton: Blackwell; 2006.

6. Royal Australian and New Zealand College of Obstetricians and Gynaecologists (RANZCOG). Vitamin and mineral supplementation in pregnancy. RANZCOG College Statement: C-Obs 25 2013. Available from URL: <http://www.ranzcog.edu.au/college-statements-guidelines.html>

9. Glossary

Definitions of anaemia are different in pregnant women compared with non-pregnant women. The World Health Organization (WHO) define anaemia in pregnancy as follows¹:

Term	Definition
Anaemia in pregnancy	First trimester – Hemoglobin <110 g/dL (approximately equivalent to a hematocrit <33 percent) Second trimester – Hemoglobin <105 g/dL (approximate hematocrit <31 or 32 percent) Third trimester – Hemoglobin level <110 g/dL (approximate hematocrit <33 percent) Post-Partum - <100 g/l

10. Document owner

Head of Department Obstetrics and Gynaecology

Enquiries relating to this guideline may be directed to:

Title: Head of Department
 Department: Obstetrics and Gynaecology
 Email: Sangeeta.MallaBhat@health.wa.gov.au

11. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

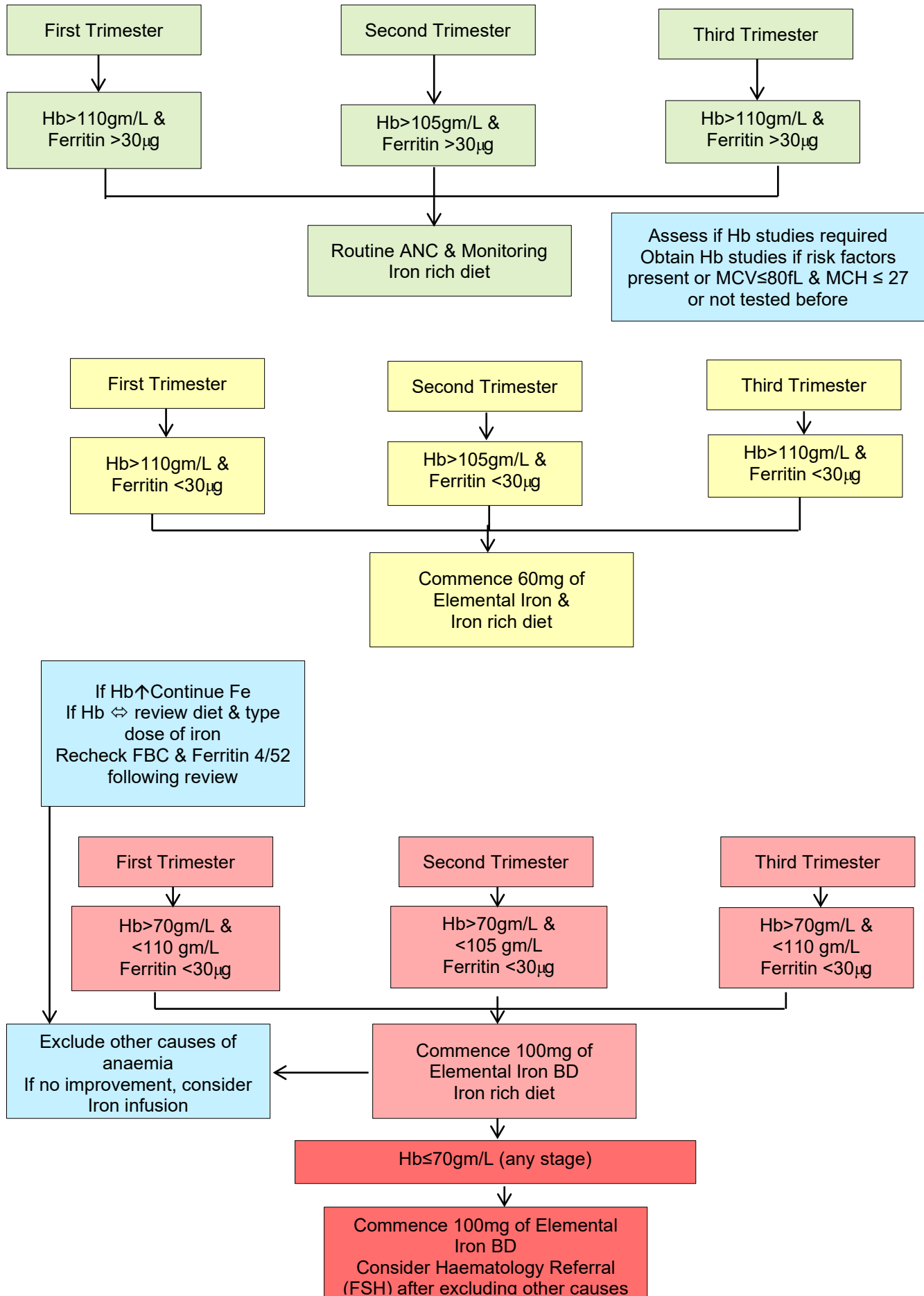
Version	Effective from	Effective to	Amendment(s)
V3.0	15/04/2025	15/04/2028	Scheduled review

12. Authorisation

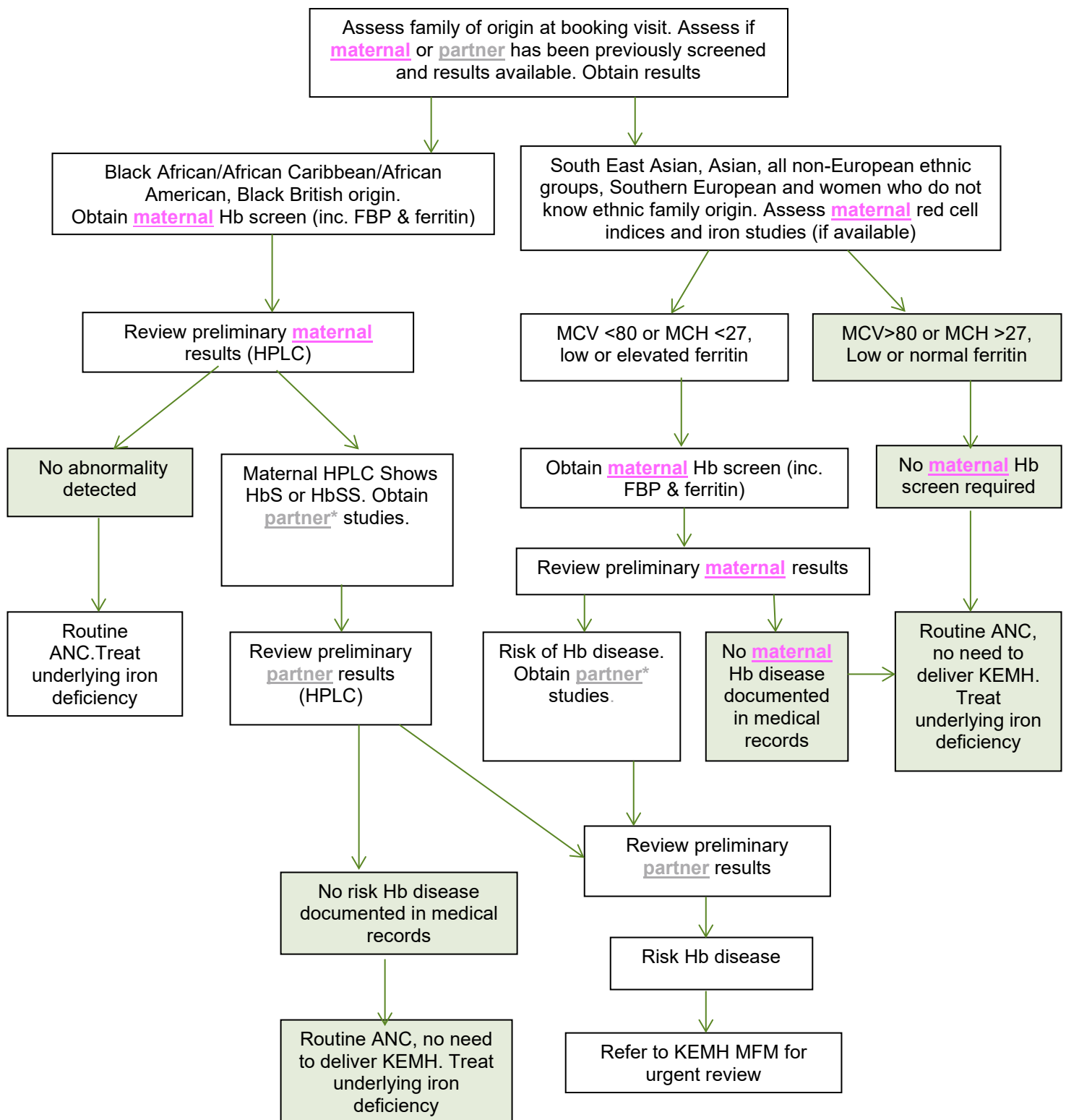
This guideline has been authorised and issued by the Director of Clinical Services.

Authorised by	Dr Tania Strickland – Director of Clinical Services
Authorisation date	09/04/2025
Published date	15/04/2025

Appendix 1: Anaemia in Pregnancy



Appendix2: Screening Women for Haemoglobin Disorders: Flow chart



*If partner declines testing or unknown, refer patient to KEMH MFM for urgent review